

PDF similarity is not one problem

Sixteen methodologies tested on the same five controlled document failures

REAL PILOT · NATIVE SCORES · NO COMPOSITE

Moderate Tables

Source-backed grouped table sample

1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table 1: 2512.03369_table_1

Dataset	Coverage	Task	Spa. Reso.	Temp. Reso.	Area (km²)	Period	Fig./Video	Device	Mod.
Sm2Boat-Fire	Worldwide	Fire Mask Forecast	30 m	1 hour	20,000,000	2013-2023	Fig	Satellite	5
FireSpeedTS	USA	Fire Mask Backtrack	375 m	1 day	9,834,000	2018-2021	Fig	Satellite	4
Mesogreco	Mediterranean	Fire Mask Forecast	1000 m	1 day	9,000,999	2006-2022	Fig	Satellite	4
FireDB	USA	Fire Intensity Forecast	375 m	1 day	9,834,000	2012-2017	Fig	Satellite	4
Next Day fire	USA	Fire Mask Forecast	1 km	1 day	9,834,000	2012-2020	Fig	Satellite	4
SeaFire Cuba	Worldwide	Fire Behavior	500 m	1 day	149,000,000	2001-2017	Fig	Satellite	4
CFDSB	Canada	Fire Behavior	180 m	1 day	9,985,000	2002-2021	Fig	Satellite	4
BA-CNFIRE	Worldwide	Fire Behavior	111 km	1 month	149,000,000	1950-2021	Fig	Satellite	1
GlobalFire	Worldwide	Fire Behavior	500 m	1 day	149,000,000	2001-2017	Fig	Satellite	1
Be-FWI	Canada	Fire Behavior	80 km	1 day	9,985,000	1980-2018	Fig	Satellite	7
GFRE	Worldwide	Fire Behavior	20 m	16 days	149,000,000	2003-2016	Fig	Satellite	1
CFDS-2005	Worldwide	Fire Behavior	1 km	16 days	149,000,000	2002-2005	Fig	Satellite	1
GFED5	Worldwide	Fire Behavior	500 m	1 month	149,000,000	1997-2020	Fig	Satellite	1
CaCAB	Worldwide	Fire Behavior	27.2 km	1 month	149,000,000	2002-2020	Fig	Satellite	3
PT-FireSped	Portugal	Fire Behavior	4 km	14 hours	92,150	2015-2021	Fig	Satellite	1
NSRCC-HS	China	Fire Mask Forecast	3 km	1 hour	—	—	—	—	—
ECUP	Regions in South Korea	Fire Monitoring	1 km	1 month	—	—	—	—	—
ECUP	Regions in South Korea	Fire Monitoring	1 km	1 month	665	2017-2021	Fig	Satellite	5
LCLQ	Regions in China	Fire Monitoring	500 m	12 hour	16,411	2015-2020	Fig	Satellite	4
CalStaV	Regions in USA	Fire Mask Forecast	20 m	1 year	450,000	2015-2022	Fig	Satellite	1
CABAM	Worldwide	Fire Behavior	30 m	1 year	149,000,000	1998-2021	Fig	Satellite	1
Fire Atlas	Worldwide	Fire Behavior	500 m	1 day	149,000,000	2003-2016	Fig	Satellite	1
MODIS	Worldwide	Fire Mask Forecast	1 km	1 day	149,000,000	2000-2024	Fig	Satellite	3
VIIRS	Worldwide	Fire Mask Forecast	375 m	12 hours	149,000,000	2012-2024	Fig	Satellite	3
NOAA IRIS Fire	North America	Danger Forecast	2 km	1 day	21,710,000	2003-2024	Fig	Satellite	3
NOAA IRIS Smoke	North America	Danger Forecast	2 km	1 day	21,710,000	2003-2024	Fig	Satellite	1
GOES fire	Western Hemisphere	Fire Mask Forecast	2 km	5 mins	61,000,000	2017-2024	Fig	Satellite	4
NIFC WP	USA	Danger Forecast	2 km	5 mins	9,834,000	2000-2024	Fig	Satellite	1
FRV	Worldwide	Fire Mask Forecast	500 m	1 day	149,000,000	2005-2011	Fig	Satellite	1
FireSentry	Regions in China	Fire Video Prediction Infrared Video Prediction	0.5 m	1 s	0.7	2025.2	Video	Drone	4

1

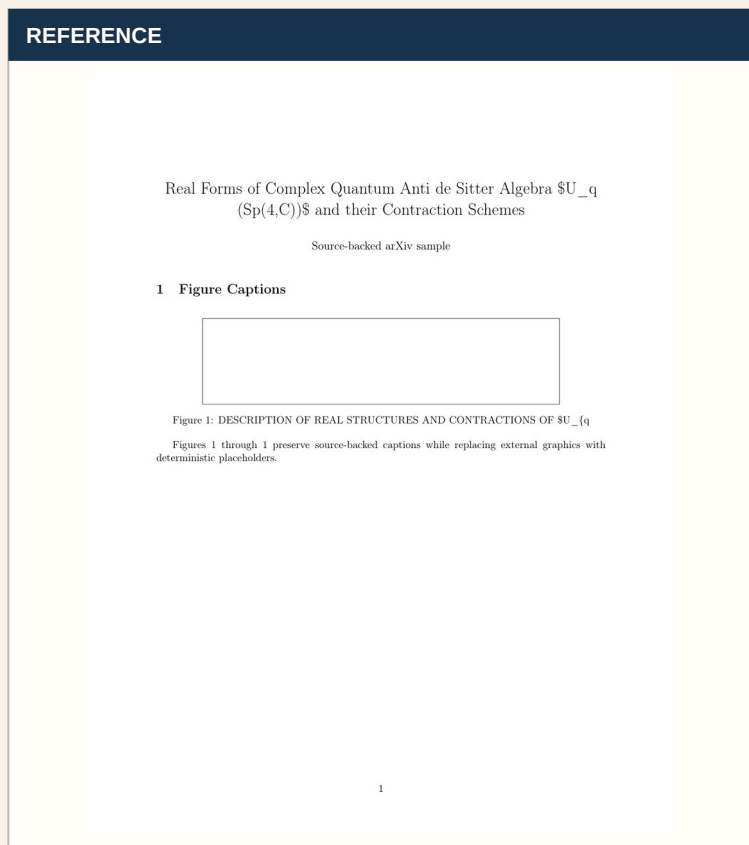
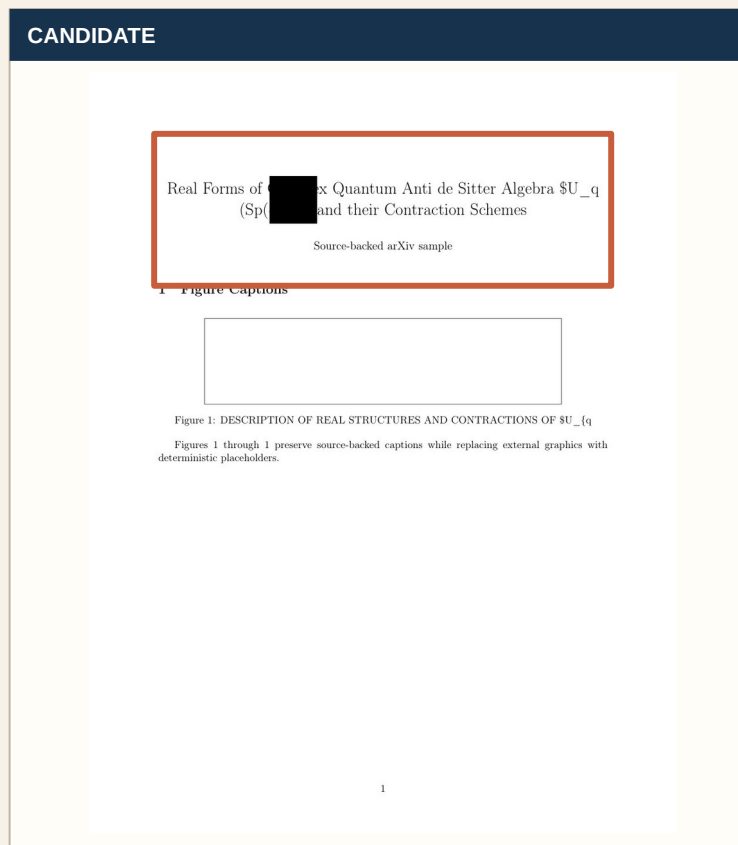
A high score can still describe a wrong PDF

0.994

SSIM for a page with a black block across its title

1.000

Token F1 for a document with three extra blank pages

REFERENCE	CANDIDATE
	

The problem is not that these metrics are bad. They answer narrower questions than “Is this PDF faithful?”

Every method sees the same five controlled failures

One severe but interpretable perturbation per content type.

The image displays five examples of document rendering issues, labeled E1 through E5, each showing a snippet of a document with a specific problem highlighted.

- E1 Prose:** A snippet of text from a paper titled "A SU(N) Gauge Theory in Three Dimensions as an Ensemble of Surfaces". The text is heavily blurred, making it illegible.
- E2 Mathematics:** A snippet of text from a paper titled "Aligned Mathematical Structures". It contains mathematical expressions, but the symbols and formulas are blurred.
- E3 Table:** A snippet of text from a paper titled "Moderate Tables". It contains a table with multiple columns and rows, but the content is blurred.
- E4 Figure:** A snippet of text from a paper titled "Real Forms of Quantum Anti-de Sitter Algebras SU_q and their Contractions". It contains a figure, but the content is blurred.
- E5 Multi-page paper:** A snippet of text from a paper titled "Loop Equations and the Topological Phase of Multi-Cut Matrix Models". It contains a figure, but the content is blurred.

This is a method-development suite—not a claim about production frequency or human severity.

These are native metric outputs—not fidelity percentages

- | | | |
|-----------|------------------------------|--|
| 01 | Direction is explicit | Every slide states whether higher or lower is better. |
| <hr/> | | |
| 02 | Bars are relative | Bar length only shows response within these five examples. |
| <hr/> | | |
| 03 | Scores stay native | Decibels remain decibels; error rates remain error rates. |
| <hr/> | | |
| 04 | No overall score | Nothing is weighted or human-calibrated in this pilot. |
| <hr/> | | |

The question on every evidence slide: what failure does this method notice, and what does it excuse?

Pixel error measures change, not importance

Average the absolute gray-level difference at every aligned pixel.

MAE = mean |R - C|

MEASURES

Total changed pixel intensity

USEFUL WHEN

Exact raster regression and rendering bugs

BLIND SPOT

Sparse but critical errors occupy too little page area

E3 · Table

[illegible]

1000 px grayscale pages; same-index target page

MAE punishes movement more than missing content

MAE · lower is better

REFERENCE

Moderate Tables

Source-backed grouped table sample

1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table 1: 2512.03369 table 1

Dataset	Coverage	Task	Sp. Mem.	Temp. Mem.	Access. Mem.	Period	Fig/Video	Device	Mod.
SmallRef-Fin	Worldwide	Fin. Mark. Forecast	30 m	1 hour	20,000,000	2013-2023	Fig.	Sanitize	5
FinRef-Fin	USA	Fin. Mark. Forecast	375 m	1 day	9,834,000	2018-2023	Fig.	Sanitize	4
Mongox	Mediterranean	Fin. Mark. Forecast	1000 m	1 day	9,000,000	2006-2022	Fig.	Sanitize	4
FinRef	USA	Fin. Mark. Forecast	375 m	1 day	9,834,000	2013-2023	Fig.	Sanitize	4
SmallRef-Fin	Worldwide	Fin. Mark. Forecast	30 m	1 hour	20,000,000	2013-2023	Fig.	Sanitize	5
CPUS	Canada	Fin. Mark. Forecast	180 m	1 day	9,834,000	2007-2023	Fig.	Sanitize	4
USANVIR	Worldwide	Fin. Mark. Forecast	113 m	1 month	10,000,000	2012-2023	Fig.	Sanitize	1
GARef	Worldwide	Fin. Mark. Forecast	500 m	1 day	10,000,000	2007-2023	Fig.	Sanitize	1
BigFUT	Canada	Fin. Mark. Forecast	80 km	1 day	9,834,000	1989-2023	Fig.	Sanitize	4
CPUS	Worldwide	Fin. Mark. Forecast	80 m	10 days	10,000,000	2013-2026	Fig.	Sanitize	1
FIN-CPUS	USA	Fin. Mark. Forecast	1 km	1 day	9,834,000	1993-2020	Fig.	Sanitize	4
CPUS	Worldwide	Fin. Mark. Forecast	500 m	1 month	10,000,000	1987-2020	Fig.	Sanitize	1
CPUS	Worldwide	Fin. Mark. Forecast	277 km	1 month	10,000,000	2002-2020	Fig.	Sanitize	3
PT-FinRef	Portugal	Fin. Mark. Forecast	4 km	14 hours	92,130	2015-2023	Fig.	Sanitize	1
SSA-CR	China	Fin. Mark. Forecast	1 km	1 hour	9,357,000	2019-2023	Fig.	Sanitize	1
WU-CR	Western USA	Fin. Mark. Forecast	200 m	3 weeks	2,200,000	2005-2017	Fig.	Sanitize	1
ECUP	Region in South Korea	Commodity Prediction	1 km	1 month	605	2017-2023	Fig.	Sanitize	5
LCUQ	Region in China	Commodity Prediction	500 m	12 hours	16,411	2015-2020	Fig.	Sanitize	4
CalRef	Region in USA	Fin. Mark. Forecast	20 m	1 year	450,000	2015-2022	Fig.	Sanitize	1
GADAM	Worldwide	Fin. Mark. Forecast	30 m	1 year	10,000,000	1998-2023	Fig.	Sanitize	1
Fin. Ref	Worldwide	Fin. Mark. Forecast	500 m	1 day	10,000,000	2003-2020	Fig.	Sanitize	1
MODS	Worldwide	Fin. Mark. Forecast	1 km	1 day	10,000,000	2009-2024	Fig.	Sanitize	3
VIRIS	Worldwide	Image Forecast	375 m	12 hours	10,000,000	2012-2024	Fig.	Sanitize	3
NOAA-HRIF-Fin	North America	Image Forecast	7 km	1 day	10,710,000	2003-2024	Fig.	Sanitize	3
NOAA-HRIF-Sat	North America	Image Forecast	7 km	1 day	10,710,000	2003-2024	Fig.	Sanitize	1
GDH-Sat	Western Hemisphere	Image Forecast	2 km	3 weeks	42,000,000	2017-2024	Fig.	Sanitize	4
SWP-WP	USA	Image Forecast	2 km	3 weeks	42,000,000	2017-2024	Fig.	Sanitize	1
FWT	Worldwide	Image Forecast	500 m	1 day	10,000,000	2003-2023	Fig.	Sanitize	1
Fin-CPUS	Canada	Fin. Mark. Forecast	1 km	1 day	9,834,000	2013-2023	Fig.	Sanitize	3
FinRef-Fin	Region in China	Fin. Mark. Forecast	0.5 m	1 s	0.7	2021-2	Video	Debase	4

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CANDIDATE

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Table 1: Source table 1: 2512.03369 table 1

Dataset	Coverage	Task	Sp. Mem.	Temp. Mem.	Access. Mem.	Period	Fig/Video	Device	Mod.
SmallRef-Fin	Worldwide	Fin. Mark. Forecast	30 m	1 hour	20,000,000	2013-2023	Fig.	Sanitize	5
FinRef-Fin	USA	Fin. Mark. Forecast	375 m	1 day	9,834,000	2018-2023	Fig.	Sanitize	4
Mongox	Mediterranean	Fin. Mark. Forecast	1000 m	1 day	9,000,000	2006-2022	Fig.	Sanitize	4
FinRef	USA	Fin. Mark. Forecast	375 m	1 day	9,834,000	2013-2023	Fig.	Sanitize	4
SmallRef-Fin	Worldwide	Fin. Mark. Forecast	30 m	1 hour	20,000,000	2013-2023	Fig.	Sanitize	5
CPUS	Canada	Fin. Mark. Forecast	180 m	1 day	9,834,000	2007-2023	Fig.	Sanitize	4
USANVIR	Worldwide	Fin. Mark. Forecast	113 m	1 month	10,000,000	2012-2023	Fig.	Sanitize	1
GARef	Worldwide	Fin. Mark. Forecast	500 m	1 day	10,000,000	2007-2023	Fig.	Sanitize	1
BigFUT	Canada	Fin. Mark. Forecast	80 km	1 day	9,834,000	1989-2023	Fig.	Sanitize	4
CPUS	Worldwide	Fin. Mark. Forecast	80 m	10 days	10,000,000	2013-2026	Fig.	Sanitize	1
FIN-CPUS	USA	Fin. Mark. Forecast	1 km	1 day	9,834,000	1993-2020	Fig.	Sanitize	4
CPUS	Worldwide	Fin. Mark. Forecast	500 m	1 month	10,000,000	1987-2020	Fig.	Sanitize	1
CPUS	Worldwide	Fin. Mark. Forecast	277 km	1 month	10,000,000	2002-2020	Fig.	Sanitize	3
PT-FinRef	Portugal	Fin. Mark. Forecast	4 km	14 hours	92,130	2015-2023	Fig.	Sanitize	1
SSA-CR	China	Fin. Mark. Forecast	1 km	1 hour	9,357,000	2019-2023	Fig.	Sanitize	1
WU-CR	Western USA	Fin. Mark. Forecast	200 m	3 weeks	2,200,000	2005-2017	Fig.	Sanitize	1
ECUP	Region in South Korea	Commodity Prediction	1 km	1 month	605	2017-2023	Fig.	Sanitize	5
LCUQ	Region in China	Commodity Prediction	500 m	12 hours	16,411	2015-2020	Fig.	Sanitize	4
CalRef	Region in USA	Fin. Mark. Forecast	20 m	1 year	450,000	2015-2022	Fig.	Sanitize	1
GADAM	Worldwide	Fin. Mark. Forecast	30 m	1 year	10,000,000	1998-2023	Fig.	Sanitize	1
Fin. Ref	Worldwide	Fin. Mark. Forecast	500 m	1 day	10,000,000	2003-2020	Fig.	Sanitize	1
MODS	Worldwide	Fin. Mark. Forecast	1 km	1 day	10,000,000	2009-2024	Fig.	Sanitize	3
VIRIS	Worldwide	Image Forecast	375 m	12 hours	10,000,000	2012-2024	Fig.	Sanitize	3
NOAA-HRIF-Fin	North America	Image Forecast	7 km	1 day	10,710,000	2003-2024	Fig.	Sanitize	3
NOAA-HRIF-Sat	North America	Image Forecast	7 km	1 day	10,710,000	2003-2024	Fig.	Sanitize	1
GDH-Sat	Western Hemisphere	Image Forecast	2 km	3 weeks	42,000,000	2017-2024	Fig.	Sanitize	4
SWP-WP	USA	Image Forecast	2 km	3 weeks	42,000,000	2017-2024	Fig.	Sanitize	1
FWT	Worldwide	Image Forecast	500 m	1 day	10,000,000	2003-2023	Fig.	Sanitize	1
Fin-CPUS	Canada	Fin. Mark. Forecast	1 km	1 day	9,834,000	2013-2023	Fig.	Sanitize	3
FinRef-Fin	Region in China	Fin. Mark. Forecast	0.5 m	1 s	0.7	2021-2	Video	Debase	4

1

FIVE CONTROLLED OUTPUTS

bar = relative penalty response within this pilot

E1

Prose shift

8.37

E2

Math blur

3.94

E3

Table deletion

0.34

E4

Figure block

1.21

E5

Extra pages

0.09

A nine-pixel page shift scores 25× worse than deleting 20 table entries. Area dominates meaning.

PSNR turns squared pixel error into decibels

Compare peak signal power with mean squared reconstruction error.

$$\text{PSNR} = 10 \log_{10}(255^2 / \text{MSE})$$

MEASURES

Pixel fidelity on a logarithmic scale

USEFUL WHEN

Compression and deterministic image pipelines

BLIND SPOT

High values can coexist with missing words or pages

E3 · Table

Moderate Tables

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Table 1: Source table 1: 2512.03369 table 1										
Dataset	Coverage	Task	Run Time	Run Time (sec)	Period	Fig./Video	Device	Mod.		
Satellite Pic	Worldwide	For Mark Forecast	20 m	1 hour	20,000,000	2018-2021	Fig.	Satellite	5	
		For Mark Backtrack	20 m	1 hour	20,000,000	2018-2021	Fig.	Satellite	5	
FingerprintTS	USA	For Mark Forecast	20 m	1 hour	5,000,000	2018-2021	Fig.	Satellite	4	
		For Mark Backtrack	20 m	1 hour	5,000,000	2018-2021	Fig.	Satellite	4	
Milestone	Middle-eastern	For Mark Forecast	100 m	1 day	5,000,000	2008-2022	Fig.	Satellite	4	
		For Mark Backtrack	100 m	1 day	5,000,000	2008-2022	Fig.	Satellite	4	
Fut01	USA	For Mark Forecast	20 m	1 day	5,000,000	2012-2017	Fig.	Satellite	4	
		For Mark Backtrack	20 m	1 day	5,000,000	2012-2017	Fig.	Satellite	4	
Sea View	USA	For Mark Forecast	10 m	1 day	5,000,000	2012-2017	Fig.	Satellite	4	
		For Mark Backtrack	10 m	1 day	5,000,000	2012-2017	Fig.	Satellite	4	
SeaView Color	Worldwide	For Mark Forecast	100 m	1 day	10,000,000	2012-2017	Fig.	Satellite	4	
		For Mark Backtrack	100 m	1 day	10,000,000	2012-2017	Fig.	Satellite	4	
VSI	Canada	For Mark Forecast	100 m	1 day	5,000,000	2007-2012	Fig.	Satellite	4	
		For Mark Backtrack	100 m	1 day	5,000,000	2007-2012	Fig.	Satellite	4	
BL-CONVSE	Worldwide	For Mark Forecast	111 km	1 month	10,000,000	1950-2017	Fig.	Satellite	1	
		For Mark Backtrack	111 km	1 month	10,000,000	1950-2017	Fig.	Satellite	1	
Globe	Canada	For Mark Forecast	100 m	1 day	5,000,000	2007-2012	Fig.	Satellite	4	
		For Mark Backtrack	100 m	1 day	5,000,000	2007-2012	Fig.	Satellite	4	
In-Pol	Canada	For Mark Forecast	100 km	1 day	5,000,000	1993-2018	Fig.	Satellite	1	
		For Mark Backtrack	100 km	1 day	5,000,000	1993-2018	Fig.	Satellite	1	
FPA FUS-A	USA	For Mark Forecast	1 km	1 day	5,000,000	1992-2020	Fig.	Satellite	4	
		For Mark Backtrack	1 km	1 day	5,000,000	1992-2020	Fig.	Satellite	4	
LQFI	Worldwide	For Mark Forecast	100 m	1 month	10,000,000	1997-2020	Fig.	Satellite	1	
		For Mark Backtrack	100 m	1 month	10,000,000	1997-2020	Fig.	Satellite	1	
LQFI	Worldwide	For Mark Forecast	100 m	1 month	10,000,000	1997-2020	Fig.	Satellite	1	
		For Mark Backtrack	100 m	1 month	10,000,000	1997-2020	Fig.	Satellite	1	
PT Fingerprint	Thailand	For Mark Forecast	1 km	14 hours	92,000	2015-2021	Fig.	Satellite	1	
		For Mark Backtrack	1 km	14 hours	92,000	2015-2021	Fig.	Satellite	1	
SHS-DB	China	For Mark Forecast	1 km	1 hour	—	—	—	—	—	
		For Mark Backtrack	1 km	1 hour	—	—	—	—	—	
RCVP	Region in South Korea	Forecast	1 km	1 month	605	2017-2021	Fig.	Satellite	5	
		Backtrack	1 km	1 month	605	2017-2021	Fig.	Satellite	5	
LC2Q	Region in China	Forecast	100 m	12 hours	56,411	2015-2020	Fig.	Satellite	4	
		Backtrack	100 m	12 hours	56,411	2015-2020	Fig.	Satellite	4	
CABM	Region in USA	Forecast	20 m	1 hour	40,000	2012-2021	Fig.	Satellite	1	
		Backtrack	20 m	1 hour	40,000	2012-2021	Fig.	Satellite	1	
CABM	Worldwide	Forecast	20 m	1 hour	10,000,000	1996-2021	Fig.	Satellite	1	
		Backtrack	20 m	1 hour	10,000,000	1996-2021	Fig.	Satellite	1	
MDS	Worldwide	Forecast	1 km	1 day	10,000,000	2008-2021	Fig.	Satellite	3	
		Backtrack	1 km	1 day	10,000,000	2008-2021	Fig.	Satellite	3	
VMS	Worldwide	Forecast	100 m	12 hours	10,000,000	2012-2021	Fig.	Satellite	3	
		Backtrack	100 m	12 hours	10,000,000	2012-2021	Fig.	Satellite	3	
MOSA-RS Pic	North America	Forecast	2 km	1 day	2,700,000	2001-2021	Fig.	Satellite	4	
		Backtrack	2 km	1 day	2,700,000	2001-2021	Fig.	Satellite	4	
GORE Inc	Western Hemisphere	Forecast	2 km	5 years	41,000,000	2017-2024	Fig.	Satellite	4	
		Backtrack	2 km	5 years	41,000,000	2017-2024	Fig.	Satellite	4	
MUTUQ	USA	Forecast	1 km	1 hour	5,000,000	2008-2012	Fig.	Satellite	1	
		Backtrack	1 km	1 hour	5,000,000	2008-2012	Fig.	Satellite	1	
FV	Worldwide	Forecast	100 m	1 day	10,000,000	2005-2011	Fig.	Satellite	1	
		Backtrack	100 m	1 day	10,000,000	2005-2011	Fig.	Satellite	1	
FVCanada	Canada	Forecast	1 km	1 day	—	—	—	—	—	
		Backtrack	1 km	1 day	—	—	—	—	—	
FVcountry	Region in China	Forecast	0.5 km	1 s	0.7	2025.2	Video	Drone	4	
		Backtrack	0.5 km	1 s	0.7	2025.2	Video	Drone	4	

1

8-bit grayscale; target-page comparison

PSNR is readable engineering math—not document truth

PSNR · higher is better

REFERENCE

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Table 1: Source table 1: 2512.03369 table 1

Dataset	Coverage	Task	Sp. Res.	Temp. Res.	Arns. Res.	Period	Fig./Video	Device	Mod.
SmallRef-Fix	Worldwide	Fix Mask Forecast	30 m	1 hour	20,000,000	2013-2021	Fig	Sanitize	1
FullRef-Fix	USA	Fix Mask Forecast	375 m	1 day	9,834,000	2018-2021	Fig	Sanitize	4
Monogpus	Mediterranean	Fix Mask Forecast	1000 m	1 day	9,000,000	2006-2022	Fig	Sanitize	4
FinRef	USA	Fix Mask Forecast	375 m	1 day	9,834,000	2012-2021	Fig	Sanitize	4
Next-Ref-Fix	USA	Fix Mask Forecast	1 km	1 day	9,834,000	2012-2021	Fig	Sanitize	4
Small-Fix-Cov	Worldwide	Fix Mask Forecast	30 m	1 day	9,834,000	2012-2021	Fig	Sanitize	4
CPUSet	Canada	Fix Mask Forecast	100 m	1 day	9,834,000	2002-2021	Fig	Sanitize	4
ILCASVIRIL	Worldwide	Fix Mask Forecast	111 km	1 month	10,000,000	2012-2021	Fig	Sanitize	1
GARef-Fix	Worldwide	Fix Mask Forecast	500 m	1 day	10,000,000	2012-2021	Fig	Sanitize	1
Big-Fix	Canada	Fix Mask Forecast	80 km	1 day	9,834,000	1989-2021	Fig	Sanitize	1
CPUS	Worldwide	Fix Mask Forecast	80 m	10 days	10,000,000	2013-2020	Fig	Sanitize	1
FFA-CPUS	USA	Fix Mask Forecast	1 km	1 day	9,834,000	1991-2020	Fig	Sanitize	4
CPUSet	Worldwide	Fix Mask Forecast	500 m	1 month	10,000,000	1987-2020	Fig	Sanitize	1
CPUSet	Worldwide	Fix Mask Forecast	271 km	1 month	10,000,000	2012-2020	Fig	Sanitize	1
PT-FireShip	Portugal	Fix Mask Forecast	4 km	14 hours	92,130	2015-2021	Fig	Sanitize	1
SSA-CIR	China	Fix Mask Forecast	1 km	1 hour	9,834,000	2019-2021	Fig	Sanitize	1
WV-CIR	Western USA	Damage Forecast	200 m	3 weeks	1,200,000	2005-2017	Fig	Sanitize	1
ECUP	Region in South Korea	Damage Prediction	1 km	1 month	605	2017-2021	Fig	Sanitize	5
LCUQ	Region in China	Damage Forecast	500 m	12 hours	16,411	2015-2020	Fig	Sanitize	4
CalRef	Region in USA	Fix Mask Forecast	20 m	1 year	10,000,000	2015-2022	Fig	Sanitize	1
GADAM	Worldwide	Fix Mask Forecast	30 m	1 year	10,000,000	1989-2021	Fig	Sanitize	1
Fix-Alto	Worldwide	Fix Mask Forecast	500 m	1 day	10,000,000	2013-2021	Fig	Sanitize	1
MODIS	Worldwide	Fix Mask Forecast	1 km	1 day	10,000,000	2000-2021	Fig	Sanitize	3
VHRS	Worldwide	Damage Forecast	375 m	12 hours	10,000,000	2012-2021	Fig	Sanitize	3
NOAA-HRR-Fix	North America	Damage Forecast	7 km	1 day	10,700,000	2003-2017	Fig	Sanitize	3
NOAA-HRR-Scale	North America	Damage Forecast	7 km	1 day	10,700,000	2003-2017	Fig	Sanitize	1
GDH-Fix	Western Hemisphere	Fix Mask Forecast	2 km	3 weeks	10,000,000	2017-2021	Fig	Sanitize	4
SHU-WP	USA	Fix Mask Forecast	2 km	3 weeks	9,834,000	2000-2021	Fig	Sanitize	1
FWT	Worldwide	Fix Mask Forecast	500 m	1 day	10,000,000	2003-2021	Fig	Sanitize	1
FireCovRef	Canada	Fix Mask Forecast	1 km	1 year	61	2013	Fig	Sanitize	3
FireCovRef	Canada	Fix Video Prediction	1 km	1 year	61	2013	Video	Sanitize	3
FireCovRef	Region in China	Fix Video Prediction	0.5 m	1 s	0.7	2012.2	Video	Sanitize	1

CANDIDATE

Moderate Tables

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1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table 1: 2512.03369 table 1

Dataset	Coverage	Task	Sp. Res.	Temp. Res.	Arns. Res.	Period	Fig./Video	Device	Mod.
SmallRef-Fix	Worldwide	Fix Mask Forecast	30 m	1 hour	20,000,000	2013-2021	Fig	Sanitize	1
FullRef-Fix	USA	Fix Mask Forecast	375 m	1 day	9,834,000	2018-2021	Fig	Sanitize	4
Monogpus	Mediterranean	Fix Mask Forecast	1000 m	1 day	9,000,000	2006-2022	Fig	Sanitize	4
FinRef	USA	Fix Mask Forecast	375 m	1 day	9,834,000	2012-2021	Fig	Sanitize	4
Next-Ref-Fix	USA	Fix Mask Forecast	1 km	1 day	9,834,000	2012-2021	Fig	Sanitize	4
Small-Fix-Cov	Worldwide	Fix Mask Forecast	30 m	1 day	9,834,000	2012-2021	Fig	Sanitize	4
CPUSet	Canada	Fix Mask Forecast	100 m	1 day	9,834,000	2002-2021	Fig	Sanitize	4
ILCASVIRIL	Worldwide	Fix Mask Forecast	111 km	1 month	10,000,000	2012-2021	Fig	Sanitize	1
GARef-Fix	Worldwide	Fix Mask Forecast	500 m	1 day	10,000,000	2012-2021	Fig	Sanitize	1
Big-Fix	Canada	Fix Mask Forecast	80 km	1 day	9,834,000	1989-2021	Fig	Sanitize	1
CPUS	Worldwide	Fix Mask Forecast	80 m	10 days	10,000,000	2013-2020	Fig	Sanitize	1
FFA-CPUS	USA	Fix Mask Forecast	1 km	1 day	9,834,000	1991-2020	Fig	Sanitize	4
CPUSet	Worldwide	Fix Mask Forecast	500 m	1 month	10,000,000	1987-2020	Fig	Sanitize	1
CPUSet	Worldwide	Fix Mask Forecast	271 km	1 month	10,000,000	2012-2020	Fig	Sanitize	1
PT-FireShip	Portugal	Fix Mask Forecast	4 km	14 hours	92,130	2015-2021	Fig	Sanitize	1
SSA-CIR	China	Fix Mask Forecast	1 km	1 hour	9,834,000	2019-2021	Fig	Sanitize	1
WV-CIR	Western USA	Damage Forecast	200 m	3 weeks	1,200,000	2005-2017	Fig	Sanitize	1
ECUP	Region in South Korea	Damage Prediction	1 km	1 month	605	2017-2021	Fig	Sanitize	5
LCUQ	Region in China	Damage Forecast	500 m	12 hours	16,411	2015-2020	Fig	Sanitize	4
CalRef	Region in USA	Fix Mask Forecast	20 m	1 year	10,000,000	2015-2022	Fig	Sanitize	1
GADAM	Worldwide	Fix Mask Forecast	30 m	1 year	10,000,000	1989-2021	Fig	Sanitize	1
Fix-Alto	Worldwide	Fix Mask Forecast	500 m	1 day	10,000,000	2013-2021	Fig	Sanitize	1
MODIS	Worldwide	Fix Mask Forecast	1 km	1 day	10,000,000	2000-2021	Fig	Sanitize	3
VHRS	Worldwide	Damage Forecast	375 m	12 hours	10,000,000	2012-2021	Fig	Sanitize	3
NOAA-HRR-Fix	North America	Damage Forecast	7 km	1 day	10,700,000	2003-2017	Fig	Sanitize	3
NOAA-HRR-Scale	North America	Damage Forecast	7 km	1 day	10,700,000	2003-2017	Fig	Sanitize	1
GDH-Fix	Western Hemisphere	Fix Mask Forecast	2 km	3 weeks	10,000,000	2017-2021	Fig	Sanitize	4
SHU-WP	USA	Fix Mask Forecast	2 km	3 weeks	9,834,000	2000-2021	Fig	Sanitize	1
FWT	Worldwide	Fix Mask Forecast	500 m	1 day	10,000,000	2003-2021	Fig	Sanitize	1
FireCovRef	Canada	Fix Mask Forecast	1 km	1 year	61	2013	Fig	Sanitize	3
FireCovRef	Canada	Fix Video Prediction	1 km	1 year	61	2013	Video	Sanitize	3
FireCovRef	Region in China	Fix Video Prediction	0.5 m	1 s	0.7	2012.2	Video	Sanitize	1

FIVE CONTROLLED OUTPUTS

bar = relative penalty response within this pilot



Deleted table content still earns 33.2 dB; the wrong five-page document earns 57.9 dB on page one.

UQI separates luminance, contrast, and correlation

Factor image similarity into correlation, mean luminance, and contrast.

$$\text{UQI} = \text{correlation} \times \text{luminance} \times \text{contrast}$$

MEASURES

Global statistical agreement

USEFUL WHEN

Broad image distortion screening

BLIND SPOT

Sparse documents keep global statistics nearly unchanged

E4 · Figure

Real Forms of $U_q(\mathfrak{sl}(n, \mathbb{C}))$ Quantum Anti de Sitter Algebra SU_q
(Sp) and their Contraction Schemes

Source-backed arXiv sample

1 Figure Captions



Figure 1: DESCRIPTION OF REAL STRUCTURES AND CONTRACTIONS OF SU_q

Figures 1 through 1 preserve source-backed captions while replacing external graphics with deterministic placeholders.

1

Sliding-window UQI on grayscale pages

Every visibly wrong candidate looks almost perfect to UQI

UQI · higher is better

REFERENCE	CANDIDATE	FIVE CONTROLLED OUTPUTS																				
Real Forms of Complex Quantum Anti de Sitter Algebra $SU_q(Sp(4,C))$ and their Contraction Schemes Source-backed arXiv sample 1 Figure Captions  Figure 1: DESCRIPTION OF REAL STRUCTURES AND CONTRACTIONS OF $SU_q(Sp(4,C))$ Figures 1 through 1 preserve source-backed captions while replacing external graphics with deterministic placeholders. 1	Real Forms of  Quantum Anti de Sitter Algebra $SU_q(Sp(4,C))$ and their Contraction Schemes Source-backed arXiv sample 1 Figure Captions  Figure 1: DESCRIPTION OF REAL STRUCTURES AND CONTRACTIONS OF $SU_q(Sp(4,C))$ Figures 1 through 1 preserve source-backed captions while replacing external graphics with deterministic placeholders. 1	bar = relative penalty response within this pilot <table><tr><td>E1</td><td>Prose shift</td><td> </td><td>0.984</td></tr><tr><td>E2</td><td>Math blur</td><td> </td><td>0.996</td></tr><tr><td>E3</td><td>Table deletion</td><td> </td><td>1.000</td></tr><tr><td>E4</td><td>Figure block</td><td> </td><td>0.995</td></tr><tr><td>E5</td><td>Extra pages</td><td> </td><td>1.000</td></tr></table>	E1	Prose shift		0.984	E2	Math blur		0.996	E3	Table deletion		1.000	E4	Figure block		0.995	E5	Extra pages		1.000
E1	Prose shift		0.984																			
E2	Math blur		0.996																			
E3	Table deletion		1.000																			
E4	Figure block		0.995																			
E5	Extra pages		1.000																			

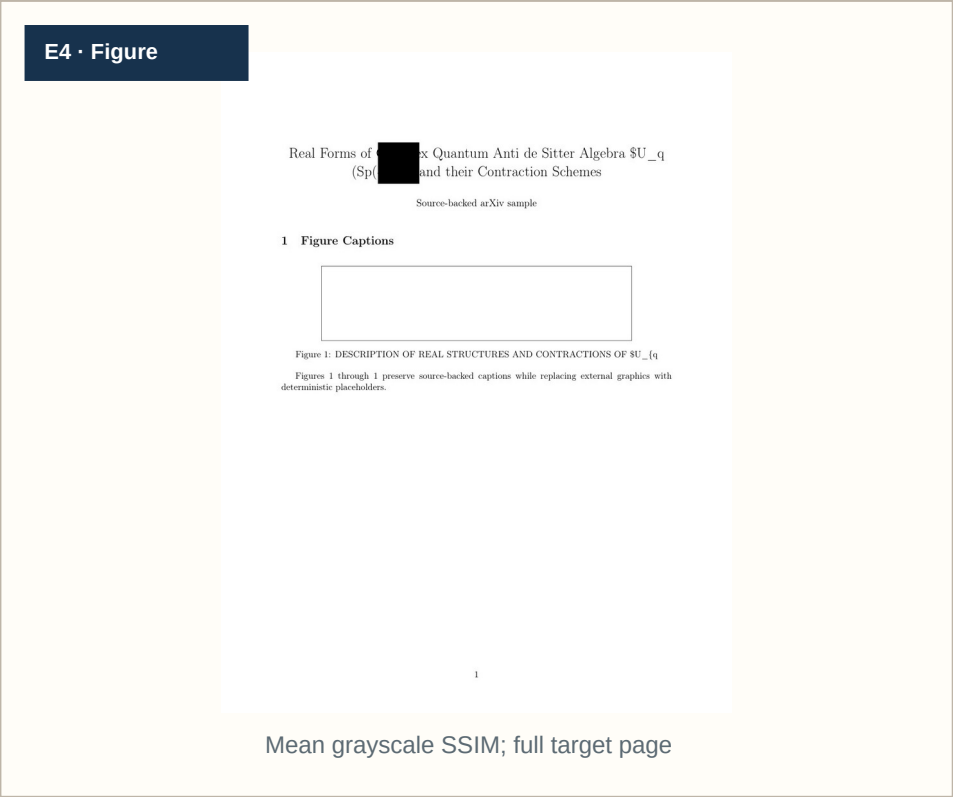
All five scores exceed 0.983—including the page with a black block over its title.

SSIM compares local structure—but assumes alignment

Compare local luminance, contrast, and structure in aligned windows.

$$\text{SSIM} = \text{luminance} \times \text{contrast} \times \text{structure}$$

MEASURES	Local structural consistency
USEFUL WHEN	Blur, compression, and broad rendering changes
BLIND SPOT	Small critical regions and document-level flow



SSIM sees translation and misses localized severity

SSIM · higher is better

REFERENCE

Real Forms of Complex Quantum Anti de Sitter Algebra SU_q
($Sp(4,C)$) and their Contraction Schemes

Source-backed arXiv sample

1 Figure Captions

Figure 1: DESCRIPTION OF REAL STRUCTURES AND CONTRACTIONS OF SU_q
Figures 1 through 1 preserve source-backed captions while replacing external graphics with deterministic placeholders.

1

CANDIDATE

Real Forms of ████████ Quantum Anti de Sitter Algebra SU_q
($Sp(████████)$) and their Contraction Schemes

Source-backed arXiv sample

1 Figure Captions

Figure 1: DESCRIPTION OF REAL STRUCTURES AND CONTRACTIONS OF SU_q
Figures 1 through 1 preserve source-backed captions while replacing external graphics with deterministic placeholders.

1

FIVE CONTROLLED OUTPUTS			bar = relative penalty response within this pilot
E1	Prose shift		0.825
E2	Math blur		0.920
E3	Table deletion		0.991
E4	Figure block		0.994
E5	Extra pages		1.000

The page shift drops to 0.825, yet a blocked title scores 0.994 and extra pages score 1.000.

Multiple scales add viewing robustness, not semantics

Aggregate structural similarity after repeated low-pass filtering and downsampling.

MS-SSIM = weighted similarity across scales

MEASURES

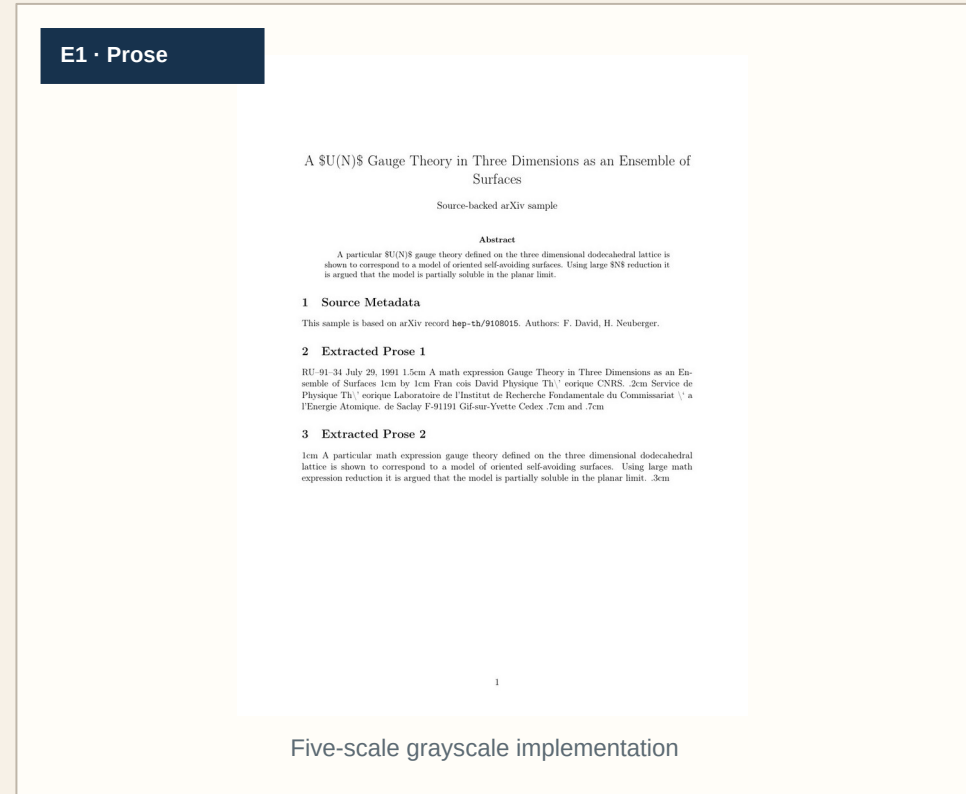
Structure visible at several resolutions

USEFUL WHEN

Large and small rendering distortions

BLIND SPOT

What the pixels mean; unmatched document pages



MS-SSIM amplifies displacement but still ignores flow

MS-SSIM · higher is better

REFERENCE

A $U(N)$ Gauge Theory in Three Dimensions as an Ensemble of Surfaces

Source-backed arXiv sample

Abstract

A particular $U(N)$ gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large N reduction it is argued that the model is partially soluble in the planar limit.

1 Source Metadata

This sample is based on arXiv record [hep-th/9108015](#). Authors: F. David, H. Neuberger.

2 Extracted Prose 1

RU-91-34 July 29, 1991 1.5cm A math expression Gauge Theory in Three Dimensions as an Ensemble of Surfaces 1cm by 1cm Fran cois David Physique Th\'eorique CNRS. 2cm Service de Physique Th\'eorique Laboratoire de l'Institut de Recherche Fondamentale du Commissariat \'\`a l'Energie Atomique, de Saclay F-91191 Gif-sur-Yvette Cedex. 7cm and 7cm

3 Extracted Prose 2

1cm A particular math expression gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large math expression reduction it is argued that the model is partially soluble in the planar limit. 3cm

1

CANDIDATE

A $U(N)$ Gauge Theory in Three Dimensions as an Ensemble of Surfaces

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A particular $U(N)$ gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large N reduction it is argued that the model is partially soluble in the planar limit.

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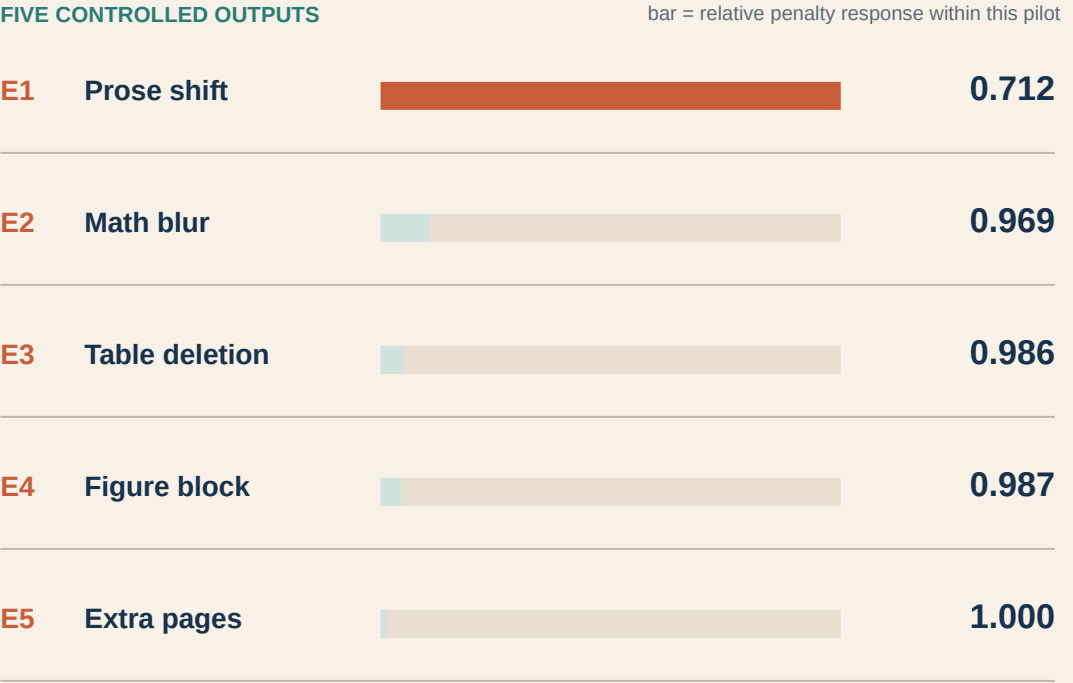
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1cm A particular math expression gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large math expression reduction it is argued that the model is partially soluble in the planar limit. 3cm

1



Translation falls to 0.712, while deletion, obstruction, and pagination remain above 0.986.

VIF estimates how much visual information survives

Estimate the information the candidate preserves relative to the reference.

VIF = candidate information /
reference information

MEASURES	Information loss under a natural-scene model
USEFUL WHEN	Blur and broad visual degradation
BLIND SPOT	Localized missing content and pagination

E2 · Mathematics

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$J_0(\delta) = \inf_{\lambda \in \mathbb{R}^+, \mu \geq 0} \left\{ (\lambda, \delta) + \sum_{k=1}^K \int \psi_k d\mu + \sum_{i=1}^L \psi_i(\alpha_i) \geq \theta_{\lambda, \mu}(\delta), \forall \delta \right\}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$C_w(x) = \begin{cases} K - vx + G_w(x + B) & \alpha_{w-1} < x \leq \tilde{\alpha}_w - B \quad k = 1, \dots, w-1 \\ \text{legmidrange} - 4pf(K - vx + G_w(\tilde{\alpha}_w)) & \tilde{\alpha}_w - B < x \leq \alpha_w \quad k = 1, \dots, w-1 \\ \text{legmidrange} - 4pf(K - vx + G_w(x + B)) & \alpha_{w-1} < x \leq \alpha_w - B \\ \text{legmidrange} - 4pf(K - vx + G_w(\alpha_w)) & \tilde{\alpha}_w - B < x \leq \alpha_w \\ \text{legmidrange} - 4pf(-vx + G_w(x)) & x \geq \alpha_w \end{cases}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$\pm \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & |n_k(n_k(E_k)) - \chi(E_k)| \\ &= |n_k(n_k(E_k)) - \chi(E_k, E_k)| \\ &\leq \int_{\mathbb{R}^n} f_n(t) \left(\int \lambda \Delta(t^{-1}a)^m v(p_k t^{-1}a, \dots, p_k t^{-1}a) - v(p_k, p_k, \dots, p_k) \right) dp_k \dots dp_m \, dt \\ &\leq \epsilon \int_{\mathbb{R}^n} f_n(t) \\ &\leq \epsilon. \end{aligned}$$

1

Pixel-domain multi-scale VIF on grayscale

VIF is powerful for rendering loss—and weak for sparse errors

VIF · higher is better

REFERENCE	CANDIDATE	FIVE CONTROLLED OUTPUTS	bar = relative penalty response within this pilot
Aligned Mathematical Structures Source-backed grouped formula sample 1 Expressions Expression 1. The following expression is taken from a source-backed formula corpus. $\mathcal{J}_D(\delta) = \inf_{\substack{\lambda \in \mathbb{R}_+^L, \psi_i > -\infty \\ (v_i)_{i \in L} \in \prod_{i \in L} L^2(\mu)}} \left\{ \langle \lambda, \delta \rangle + \sum_{i=1}^L \int \psi_i d\mu : \sum_{i=1}^L \psi_i(s_i) \geq g_{\lambda, D}(s), \forall s \right\}.$ Expression 2. The following expression is taken from a source-backed formula corpus. $C_w(x) = \begin{cases} K - vx + G_w(x+B) & s_{k-1} < x \leq \tilde{S}_k - B \quad k = 1, \dots, w-1 \\ \text{beginaligns} - 4pf[K - vx + G_w(\tilde{S}_k)] & \tilde{S}_k - B < x \leq s_k \quad k = 1, \dots, w-1 \\ \text{beginaligns} - 4pf[K - vx + G_w(x+B)] & s_{w-1} < x \leq \tilde{S}_w - B \\ \text{beginaligns} - 4pf[K - vx + G_w(S_w)] & \tilde{S}_w - B < x \leq s_w \\ \text{beginaligns} - 4pf[-vx + G_w(x)] & x > s_w. \end{cases}$ Expression 3. The following expression is taken from a source-backed formula corpus. $\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$\pm \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ Expression 4. The following expression is taken from a source-backed formula corpus. $\begin{aligned} & \phi_n(\alpha_n(E_q)) - \chi(E_q) \\ &= \phi_n(\alpha_n(E_q)) - \chi(E_q, E_q) \\ &\leq \int_{\text{set}_n} f_n(b) \left(\int \Delta(b^{-1}a) ^m \psi(g_1 b^{-1}a, \dots, g_m b^{-1}a) - \psi(g_1, g_2, \dots, g_m) \right) dg_1 \cdots dg_m \, db \\ &\leq \epsilon \int_{\text{set}_n} f_n(b) \\ &\leq \epsilon. \end{aligned}$ 1	Aligned Mathematical Structures Source-backed grouped formula sample 1 Expressions Expression 1. The following expression is taken from a source-backed formula corpus. $\mathcal{J}_D(\delta) = \inf_{\substack{\lambda \in \mathbb{R}_+^L, \psi_i > -\infty \\ (v_i)_{i \in L} \in \prod_{i \in L} L^2(\mu)}} \left\{ \langle \lambda, \delta \rangle + \sum_{i=1}^L \int \psi_i d\mu : \sum_{i=1}^L \psi_i(s_i) \geq g_{\lambda, D}(s), \forall s \right\}.$ Expression 2. The following expression is taken from a source-backed formula corpus. $C_w(x) = \begin{cases} K - vx + G_w(x+B) & s_{k-1} < x \leq \tilde{S}_k - B \quad k = 1, \dots, w-1 \\ \text{beginaligns} - 4pf[K - vx + G_w(\tilde{S}_k)] & \tilde{S}_k - B < x \leq s_k \quad k = 1, \dots, w-1 \\ \text{beginaligns} - 4pf[K - vx + G_w(x+B)] & s_{w-1} < x \leq \tilde{S}_w - B \\ \text{beginaligns} - 4pf[K - vx + G_w(S_w)] & \tilde{S}_w - B < x \leq s_w \\ \text{beginaligns} - 4pf[-vx + G_w(x)] & x > s_w. \end{cases}$ Expression 3. The following expression is taken from a source-backed formula corpus. $\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$\pm \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ Expression 4. The following expression is taken from a source-backed formula corpus. $\begin{aligned} & \phi_n(\alpha_n(E_q)) - \chi(E_q) \\ &= \phi_n(\alpha_n(E_q)) - \chi(E_q, E_q) \\ &\leq \int_{\text{set}_n} f_n(b) \left(\int \Delta(b^{-1}a) ^m \psi(g_1 b^{-1}a, \dots, g_m b^{-1}a) - \psi(g_1, g_2, \dots, g_m) \right) dg_1 \cdots dg_m \, db \\ &\leq \epsilon \int_{\text{set}_n} f_n(b) \\ &\leq \epsilon. \end{aligned}$ 1	E1 Prose shift 0.004 E2 Math blur 0.197 E3 Table deletion 0.951 E4 Figure block 0.938 E5 Extra pages 0.987	

VIF collapses on shift (0.004) and blur (0.197), but stays above 0.937 for deletion and obstruction.

Gradient inconsistency exposes damaged strokes efficiently

Measure the variation of a pixel-wise gradient-similarity map.

GMSD = std(gradient similarity map)

MEASURES

Uneven edge degradation

USEFUL WHEN

Blur, ringing, and geometric disturbance

BLIND SPOT

Sparse semantic deletion and document length

E2 · Mathematics

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$J_0(\delta) = \inf_{\lambda \in \mathbb{R}^+, \mu \geq -\infty} \left\{ (\lambda, \delta) + \sum_{k=1}^K \int \psi_k d\mu : \sum_{k=1}^K \psi_k(u_k) \geq \theta_{\lambda, \mu}(x), \forall x \right\}.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$C_k(x) = \begin{cases} K - vx + G_k(x + B) & n_{k-1} < x \leq \tilde{S}_k - B \quad k = 1, \dots, w-1 \\ \text{beginalign} - \text{bp}[K - vx + G_k(\tilde{S}_k)] & \tilde{S}_k - B < x \leq n_k \quad k = 1, \dots, w-1 \\ \text{beginalign} - \text{bp}[K - vx + G_k(x + B)] & n_{w-1} < x \leq S_{w-1} - B \\ \text{beginalign} - \text{bp}[K - vx + G_k(\tilde{S}_w)] & \tilde{S}_w - B < x \leq n_w \\ \text{beginalign} - \text{bp}[-vx + G_k(x)] & x \geq n_w \end{cases}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$\pm \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

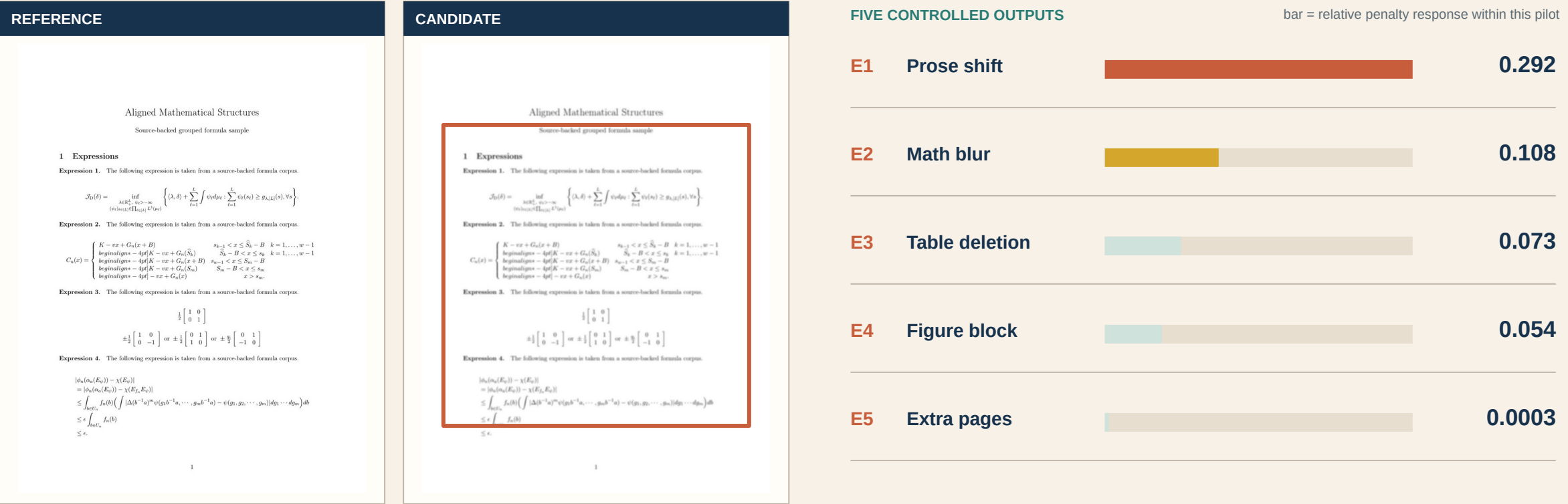
$$\begin{aligned} & |n_k(n_k(E_k)) - \chi(E_k)| \\ &= |n_k(n_k(E_k)) - \chi(E_k, E_k)| \\ &\leq \int_{\mathbb{R}^n} f_n(t) \left(\int \Delta(t^{-1}x)^n v(p_k t^{-1}x, \dots, p_w t^{-1}x) - v(p_1, p_2, \dots, p_w) \right) dx \\ &\leq \epsilon \int_{\mathbb{R}^n} f_n(t) \\ &\leq \epsilon. \end{aligned}$$

1

Original 8-bit constant T=170; downsampled gradients

GMSD responds to moved and softened edges—not extra pages

GMSD · lower is better



Binary ink overlap makes visible marks interpretable

Threshold visible ink, allow a fixed two-pixel neighborhood, then compute F1 overlap.

$$\text{Ink F1} = 2 \times \text{precision} \times \text{recall} / (\text{precision} + \text{recall})$$

MEASURES

Whether dark marks occupy the same places

USEFUL WHEN

Text displacement, missing ink, and obstructions

BLIND SPOT

Blur inside the tolerance and unseen pages

E4 · Figure

Real Forms of $Sp(2, \mathbb{C})$ Quantum Anti de Sitter Algebra SU_q
(Sp(2, C) and their Contraction Schemes)

Source-backed arXiv sample

1 Figure Captions



Figure 1: DESCRIPTION OF REAL STRUCTURES AND CONTRACTIONS OF SU_q

Figures 1 through 1 preserve source-backed captions while replacing external graphics with deterministic placeholders.

1

Threshold 220/255; 2 px symmetric dilation

A two-pixel tolerance separates movement from antialiasing

Ink F1 · higher is better

REFERENCE

Real Forms of Complex Quantum Anti de Sitter Algebra SU_q
($Sp(4,C)$) and their Contraction Schemes

Source-backed arXiv sample

1 Figure Captions



Figure 1: DESCRIPTION OF REAL STRUCTURES AND CONTRACTIONS OF SU_q
Figures 1 through 1 preserve source-backed captions while replacing external graphics with deterministic placeholders.

1

CANDIDATE

Real Forms of  Quantum Anti de Sitter Algebra SU_q
($Sp(\text{redacted})$) and their Contraction Schemes

Source-backed arXiv sample

1 Figure Captions



Figure 1: DESCRIPTION OF REAL STRUCTURES AND CONTRACTIONS OF SU_q
Figures 1 through 1 preserve source-backed captions while replacing external graphics with deterministic placeholders.

1

FIVE CONTROLLED OUTPUTS			bar = relative penalty response within this pilot
E1	Prose shift		0.312
E2	Math blur		0.996
E3	Table deletion		0.981
E4	Figure block		0.911
E5	Extra pages		1.000

It strongly penalizes shift (0.312) and obstruction (0.911), but calls blurred math almost perfect (0.996).

Chamfer distance asks how far every stroke moved

For each edge pixel, find the nearest edge in the other image and average both directions.

Chamfer = mean nearest-edge distance

MEASURES

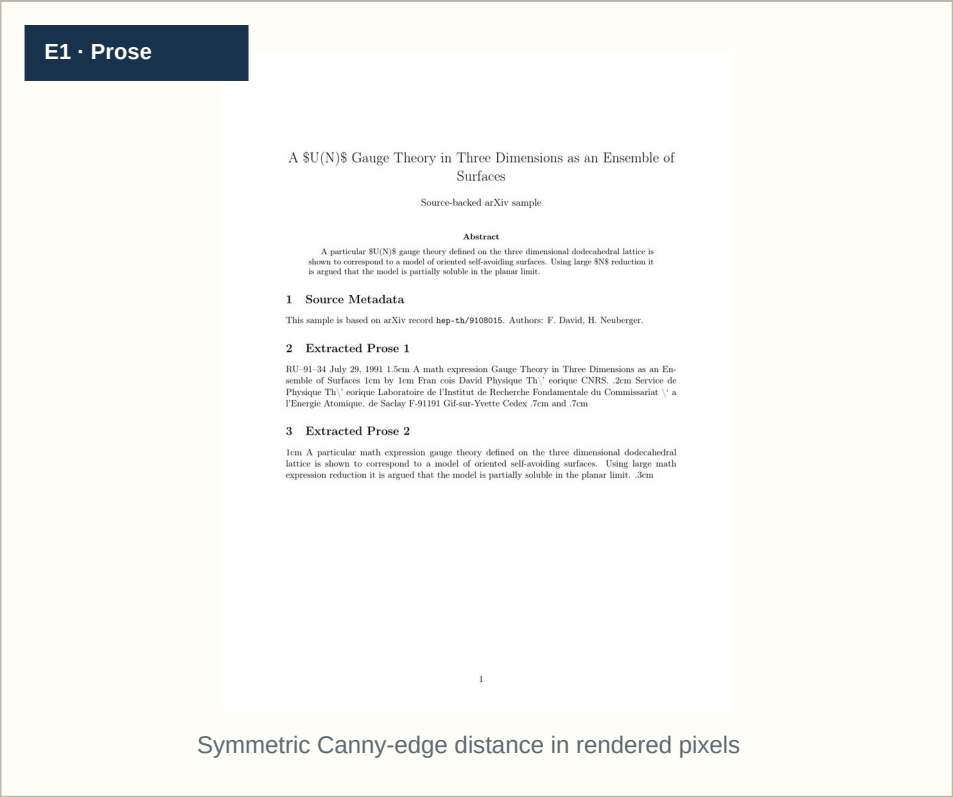
Spatial displacement of contours

USEFUL WHEN

Translation, line movement, and blur

BLIND SPOT

Small missing regions and document-level structure



Edge distance is intuitive for geometry—and blind to meaning

Chamfer · lower is better

REFERENCE

A $U(N)$ Gauge Theory in Three Dimensions as an Ensemble of Surfaces

Source-backed arXiv sample

Abstract

A particular $U(N)$ gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large N reduction it is argued that the model is partially soluble in the planar limit.

1 Source Metadata

This sample is based on arXiv record [hep-th/9108015](#). Authors: F. David, H. Neuberger.

2 Extracted Prose 1

RU-91-34 July 29, 1991 1.5cm A math expression Gauge Theory in Three Dimensions as an Ensemble of Surfaces 1cm by 1cm Fran cois David Physique Th\'eorique CNRS. 2cm Service de Physique Th\'eorique Laboratoire de l'Institut de Recherche Fondamentale du Commissariat \`a l'Energie Atomique, de Saclay F-91191 Gif-sur-Yvette Cedex. 7cm and 7cm

3 Extracted Prose 2

1cm A particular math expression gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large math expression reduction it is argued that the model is partially soluble in the planar limit. 3cm

1

CANDIDATE

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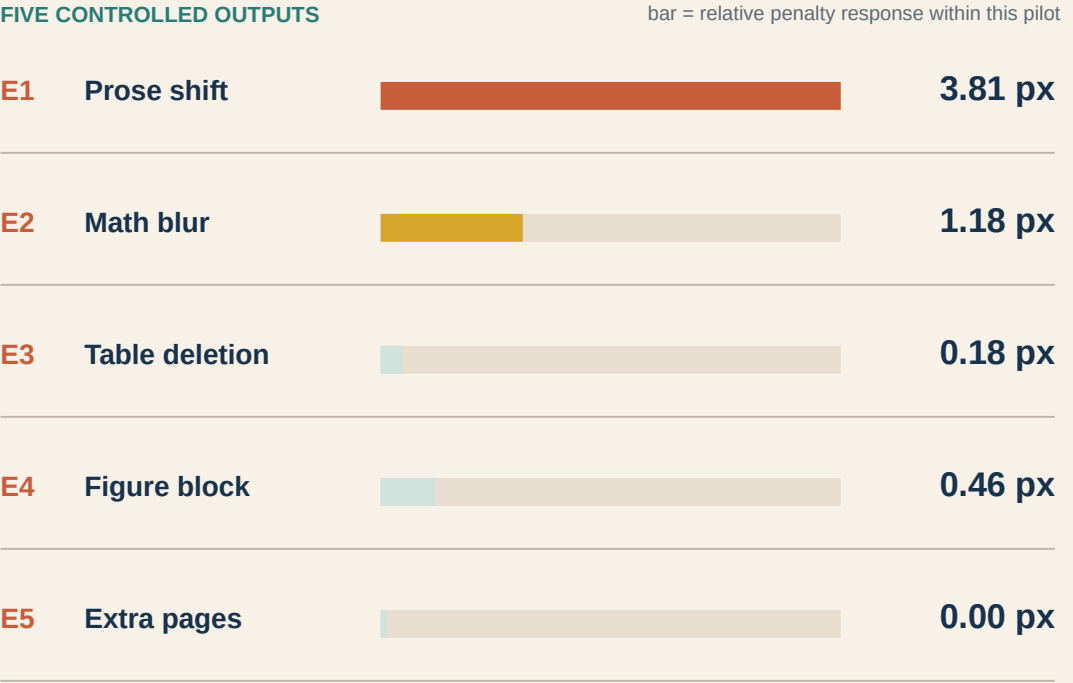
2 Extracted Prose 1

RU-91-34 July 29, 1991 1.5cm A math expression Gauge Theory in Three Dimensions as an Ensemble of Surfaces 1cm by 1cm Fran cois David Physique Th\'eorique CNRS. 2cm Service de Physique Th\'eorique Laboratoire de l'Institut de Recherche Fondamentale du Commissariat \`a l'Energie Atomique, de Saclay F-91191 Gif-sur-Yvette Cedex. 7cm and 7cm

3 Extracted Prose 2

1cm A particular math expression gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large math expression reduction it is argued that the model is partially soluble in the planar limit. 3cm

1



The nine-pixel shift produces 3.81 px average edge displacement; deleted table content produces only 0.18 px.

CER measures character edits after OCR

Align OCR character strings and count substitutions, insertions, and deletions.

**CER = character edits /
reference characters**

MEASURES

Visible lexical corruption

USEFUL WHEN

Missing text, wrong symbols, and OCR-affecting blur

BLIND SPOT

Correct text in wrong geometry; non-text objects

E3 · Table

Moderate Tables

Source-backed grouped table sample

1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table 1: 2512.03369 table 1

[illegible]

Tesseract PSM 3: NFKC and whitespace normalization

CER · lower is better

CER flags table deletion at 54.1% error, yet the blocked title is only 4.0% and extra pages are 0%.

WER makes broken and missing words expensive

Run the same edit-distance logic on words instead of characters.

$$\text{WER} = (\text{substitutions} + \text{deletions} + \text{insertions}) / \text{words}$$

MEASURES

Whole-word transcription damage

USEFUL WHEN

Missing phrases and severely blurred text

BLIND SPOT

Typography, object fidelity, geometry, and extra pages

E2 · Mathematics

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$J_0(t) = \inf_{\lambda \in \mathbb{R}, \mu \in \mathbb{R}} \left\{ (\lambda, \mu) + \sum_{k=1}^K \int \psi_k d\mu : \sum_{k=1}^K \psi_k(u) \geq \theta_{\lambda, \mu}(t), \forall t \right\}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$C_k(x) = \begin{cases} K - vx + G_k(x + B) & n_{k-1} < x \leq \tilde{S}_k - B \quad k = 1, \dots, w-1 \\ \text{legimalignes} - \text{Apf}(K - vx + G_k(\tilde{S}_k)) & \tilde{S}_k - B < x \leq n_k \quad k = 1, \dots, w-1 \\ \text{legimalignes} - \text{Apf}(K - vx + G_k(x + B)) & n_{w-1} < x \leq S_{w-1} - B \\ \text{legimalignes} - \text{Apf}(K - vx + G_k(\tilde{S}_w)) & \tilde{S}_w - B < x \leq n_w \\ \text{legimalignes} - \text{Apf}(K - vx + G_k(x)) & x > n_w \end{cases}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\pm \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{or} \quad \pm \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \text{or} \quad \pm \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \pm \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} & |n_k(n_k(E_k)) - \chi(E_k)| \\ &= |n_k(n_k(E_k)) - \chi(E_k, E_k)| \\ &\leq \int_{\mathbb{R}^n} f_n(t) \left(\int \lambda \Delta(t^{-1}a)^n v(p_k t^{-1}a, \dots, p_k t^{-1}a) - v(p_k, p_k, \dots, p_k) \right) dp_k \, dt \\ &\leq \epsilon \int_{\mathbb{R}^n} f_n(t) \\ &\leq \epsilon. \end{aligned}$$

1

Same OCR stream and normalization as CER

Word-level edits expose unusable text more aggressively

WER · lower is better

REFERENCE	CANDIDATE
Aligned Mathematical Structures Source-backed grouped formula sample 1 Expressions Expression 1. The following expression is taken from a source-backed formula corpus. $\mathcal{J}_D(\delta) = \inf_{\substack{\lambda \in \mathbb{R}_+^L, \psi: \mathcal{V} \rightarrow \mathbb{R} \\ (\psi(v_{1:2}) \in \prod_{i=1:2} L^2(\mu))}} \left\{ \langle \lambda, \delta \rangle + \sum_{i=1}^L \int \psi_i d\mu : \sum_{i=1}^L \psi_i(s_i) \geq g_{\lambda, D}(s), \forall s \right\}.$ Expression 2. The following expression is taken from a source-backed formula corpus. $C_w(x) = \begin{cases} K - vx + G_w(x+B) & s_{k-1} < x \leq \tilde{S}_k - B \quad k = 1, \dots, w-1 \\ \text{beginaligns} - 4pf[K - vx + G_w(\tilde{S}_k)] & \tilde{S}_k - B < x \leq s_k \quad k = 1, \dots, w-1 \\ \text{beginaligns} - 4pf[K - vx + G_w(x+B)] & s_{w-1} < x \leq \tilde{S}_w - B \\ \text{beginaligns} - 4pf[K - vx + G_w(S_w)] & \tilde{S}_w - B < x \leq s_w \\ \text{beginaligns} - 4pf[-vx + G_w(x)] & x > s_w. \end{cases}$ Expression 3. The following expression is taken from a source-backed formula corpus. $\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$\pm \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ Expression 4. The following expression is taken from a source-backed formula corpus. $\begin{aligned} & \phi_n(\alpha_n(E_q)) - \chi(E_q) \\ &= \phi_n(\alpha_n(E_q)) - \chi(E_q, E_q) \\ &\leq \int_{\mathcal{E} \in \mathcal{E}_n} f_n(b) \left(\int \Delta(b^{-1}a) ^m \psi(g_1 b^{-1}a, \dots, g_m b^{-1}a) - \psi(g_1, g_2, \dots, g_m) dg_1 \cdots dg_m \right) db \\ &\leq \epsilon \int_{\mathcal{E} \in \mathcal{E}_n} f_n(b) \\ &\leq \epsilon. \end{aligned}$ 1	Aligned Mathematical Structures Source-backed grouped formula sample 1 Expressions Expression 1. The following expression is taken from a source-backed formula corpus. $\mathcal{J}_D(\delta) = \inf_{\substack{\lambda \in \mathbb{R}_+^L, \psi: \mathcal{V} \rightarrow \mathbb{R} \\ (\psi(v_{1:2}) \in \prod_{i=1:2} L^2(\mu))}} \left\{ \langle \lambda, \delta \rangle + \sum_{i=1}^L \int \psi_i d\mu : \sum_{i=1}^L \psi_i(s_i) \geq g_{\lambda, D}(s), \forall s \right\}.$ Expression 2. The following expression is taken from a source-backed formula corpus. $C_w(x) = \begin{cases} K - vx + G_w(x+B) & s_{k-1} < x \leq \tilde{S}_k - B \quad k = 1, \dots, w-1 \\ \text{beginaligns} - 4pf[K - vx + G_w(\tilde{S}_k)] & \tilde{S}_k - B < x \leq s_k \quad k = 1, \dots, w-1 \\ \text{beginaligns} - 4pf[K - vx + G_w(x+B)] & s_{w-1} < x \leq \tilde{S}_w - B \\ \text{beginaligns} - 4pf[K - vx + G_w(S_w)] & \tilde{S}_w - B < x \leq s_w \\ \text{beginaligns} - 4pf[-vx + G_w(x)] & x > s_w. \end{cases}$ Expression 3. The following expression is taken from a source-backed formula corpus. $\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$\pm \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \text{ or } \pm \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ Expression 4. The following expression is taken from a source-backed formula corpus. $\begin{aligned} & \phi_n(\alpha_n(E_q)) - \chi(E_q) \\ &= \phi_n(\alpha_n(E_q)) - \chi(E_q, E_q) \\ &\leq \int_{\mathcal{E} \in \mathcal{E}_n} f_n(b) \left(\int \Delta(b^{-1}a) ^m \psi(g_1 b^{-1}a, \dots, g_m b^{-1}a) - \psi(g_1, g_2, \dots, g_m) dg_1 \cdots dg_m \right) db \\ &\leq \epsilon \int_{\mathcal{E} \in \mathcal{E}_n} f_n(b) \\ &\leq \epsilon. \end{aligned}$ 1

FIVE CONTROLLED OUTPUTS

bar = relative penalty response within this pilot

E1	Prose shift		9.0%
E2	Math blur		64.7%
E3	Table deletion		81.9%
E4	Figure block		6.7%
E5	Extra pages		0.0%

Blurred math reaches 64.7% WER and table deletion 81.9%; a wrong five-page PDF remains 0%.

Token F1 asks whether the same words survived

Compare OCR token multisets with precision, recall, and their harmonic mean.

**Token F1 = harmonic
mean(overlap precision,
overlap recall)**

MEASURES

Lexical coverage independent of order

USEFUL WHEN

Missing or substituted textual content

BLIND SPOT

Reading order, typography, geometry, and pagination

E3 • Table

Moderate Tables

Source-backed grouped table sample

1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table 1: 2512.03369_table_1

[illegible]

Lowercased Unicode word-token multiset

Bag-of-token overlap is clear—but discards order and layout

Token F1 · higher is better

REFERENCE

Moderate Tables

Source-backed grouped table sample

1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

[illegible]

1

CANDIDATE

Moderate Tables

Source-backed grouped table sample

1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table at 2512.0369, table 1									
Dataset	Coverage	Task	Spice	Time	Score	Period	File	Video	Mod.
RealEstate FSN	Worldwide	Find Mark Location	100	10	100.000	2013-2021	Fig 1	Sample	5
Facebook FSN	USA	Find Mark Location	275	1	100.000	2013-2021	Fig 2	Sample	5
Mapbox	Worldwide	Find Mark Location	100	1	98.000	2009-2022	Fig 3	Sample	4
Mapbox	Worldwide	Find Mark Location	100	1	98.000	2009-2022	Fig 4	Sample	4
New York City FSN	USA	Find Mark Location	100	1	93.000	2013-2021	Fig 5	Sample	5
OpenStreetMap	Worldwide	Find Mark Location	100	1	98.000	2013-2021	Fig 6	Sample	5
OpenStreetMap	Canada	Find Mark Location	100	1	98.000	2013-2021	Fig 7	Sample	5
Re-BUT	Canada	Find Mark Location	100	1	98.000	2013-2021	Fig 8	Sample	5
Re-BUT	Canada	Find Mark Location	100	1	93.000	1980-2013	Fig 9	Sample	5
Re-BUT	Canada	Find Mark Location	100	1	93.000	1980-2013	Fig 10	Sample	5
Re-BUT	Canada	Find Mark Location	100	1	93.000	1980-2013	Fig 11	Sample	5
Re-BUT	Canada	Find Mark Location	100	1	93.000	1980-2013	Fig 12	Sample	5
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Re-BUT	Canada	Find Mark Location	100	1	93.000	1980-2013	Fig 140	Sample	5
Re-BUT	Canada	Find Mark Location	100	1	93.000	1980-2013	Fig 141	Sample	5
Re-BUT	Canada	Find Mark Location	100	1	93.000	1980-2013	Fig 142	Sample	5
Re-BUT	Canada	Find Mark Location	10						

1

FIVE CONTROLLED OUTPUTS

bar = relative penalty response within this pilot

E1 Prose shift

0.936

E2 Math blur

0.565

E3 Table deletion

0.473

E4 Figure block

0.936

E5 Extra pages

1.000

Deleted table content falls to 0.473; the blocked figure and shifted prose remain above 0.935.

DocSim matches boxes instead of pixels

Find the maximum-weight one-to-one matching between normalized line boxes.

DocSim = maximum-weight box matching

MEASURES

Relative box position, area, and shape

USEFUL WHEN

Typed region layouts with reliable detection

BLIND SPOT

OCR segmentation instability and unmatched pages

E1 · Prose

A $U(N)$ Gauge Theory in Three Dimensions as an Ensemble of Surfaces

Source-backed arXiv sample

Abstract

A particular $SU(N)$ gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large N reduction it is argued that the model is partially soluble in the planar limit.

1 Source Metadata

This sample is based on arXiv record [hep-th/9108015](#). Authors: F. David, H. Neuberger.

2 Extracted Prose 1

RU-91-34 July 29, 1991 1.5cm A math expression Gauge Theory in Three Dimensions as an Ensemble of Surfaces 1cm by 1cm Fran cois David Physique Th\'eorique CNRS. 2cm Service de Physique Th\'eorique Laboratoire de l'Institut de Recherche Fondamentale du Commissariat \`a l'Energie Atomique, de Saclay F-91191 Gif-sur-Yvette Cedex. 7cm and. 7cm

3 Extracted Prose 2

1cm A particular math expression gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large math expression reduction it is argued that the model is partially soluble in the planar limit. .3cm

1

OCR lines; center, shape, and area weighting; no class labels

Our OCR-line adaptation collapses into a narrow score range

DocSim · higher is better

REFERENCE

A $U(N)$ Gauge Theory in Three Dimensions as an Ensemble of Surfaces

Source-backed arXiv sample

Abstract

A particular $SU(N)$ gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large N reduction it is argued that the model is partially soluble in the planar limit.

1 Source Metadata

This sample is based on arXiv record [hep-th/9108015](#). Authors: F. David, H. Neuberger.

2 Extracted Prose 1

RU-91-34 July 29, 1991 1.5cm A math expression Gauge Theory in Three Dimensions as an Ensemble of Surfaces 1cm by 1cm Fran cois David Physique Th\'eorique CNRS .2cm Service de Physique Th\'eorique Laboratoire de l'Institut de Recherche Fondamentale du Commissariat \'\`a l'Energie Atomique, de Saclay F-91191 Gif-sur-Yvette Cedex .7cm and .7cm

3 Extracted Prose 2

1cm A particular math expression gauge theory defined on the three dimensional dodecahedral lattice is shown to correspond to a model of oriented self-avoiding surfaces. Using large math expression reduction it is argued that the model is partially soluble in the planar limit. .3cm

1

CANDIDATE

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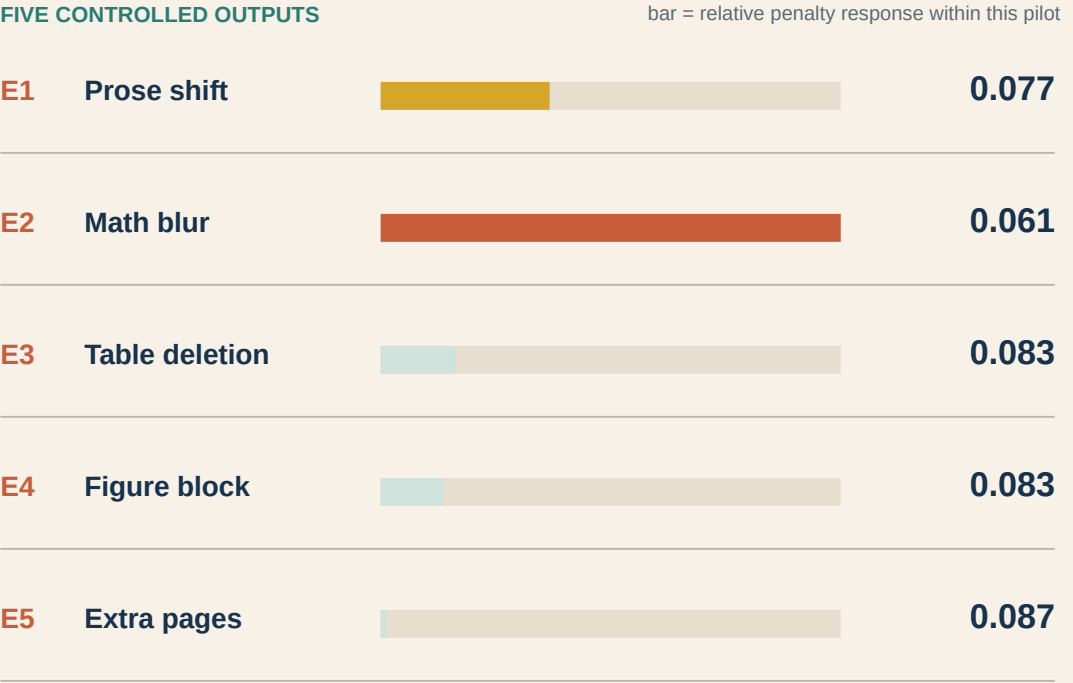
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1



All five outputs sit between 0.061 and 0.087. Without stable typed regions, the absolute score is not useful.

Text-aware IoU binds the right word to the right place

Match identical OCR tokens, compute box intersection-over-union, and penalize unmatched words.

Word IoU = $\frac{\text{matched intersection}}{\text{union}}$

MEASURES	Joint lexical and geometric agreement
USEFUL WHEN	Moved text, missing text, and reading geometry
BLIND SPOT	Figures, typography within boxes, and extra pages

E1 · Prose

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Hungarian matching of exact normalized OCR tokens

Word-box overlap exposes displacement and OCR instability

Word IoU · higher is better

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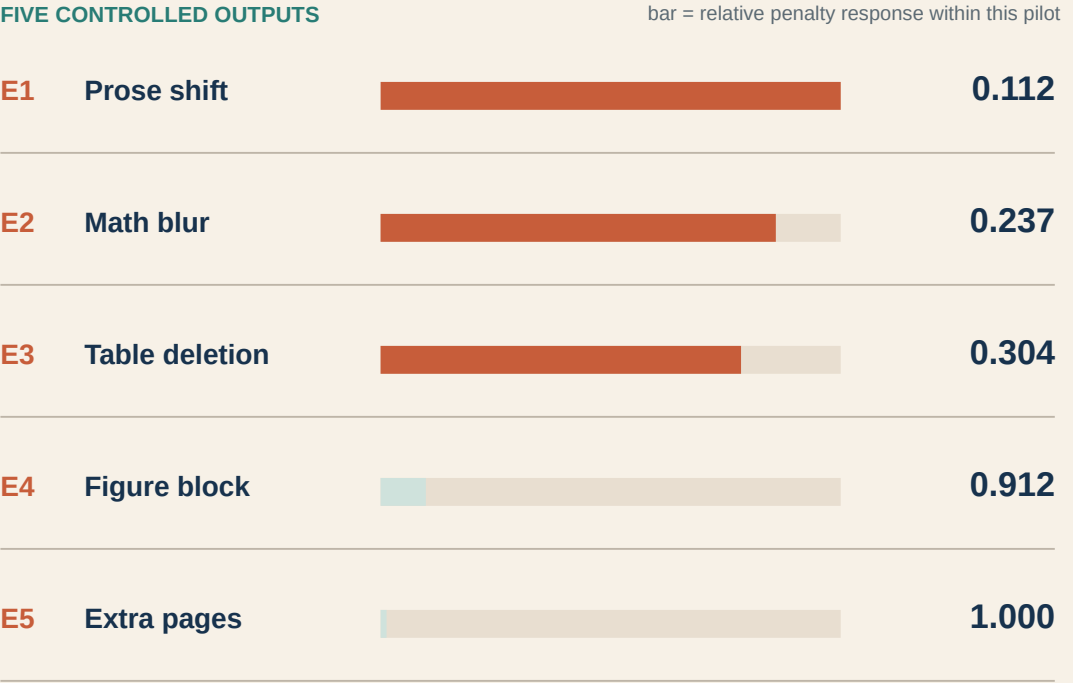
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1



The shifted page drops to 0.112 and blurred math to 0.237; a blocked non-text region still scores 0.912.

Page count is blunt—and essential

Compare document lengths before inspecting any page content.

Page ratio = min(page counts) / max(page counts)

MEASURES

Missing or extra pages

USEFUL WHEN

Fast document-integrity gating

BLIND SPOT

Everything that happens inside matched pages

E5 · Multi-page paper

Loop Equations and the Topological Phase of Multi-Cut Matrix Models

C. Crnkovic, M. Douglas, G. Moore

Abstract

We study the double scaling limit of mKdV type, realized in the two-cut Hermitian matrix model. Building on the work of Perini and Shevitz and of Nappi, we find an exact solution including all odd scaling operators, in terms of a hierarchy of flows of 82 times 28 matrices. We derive from it loop equations which can be expressed as Virasoro constraints on the partition function. We discover a "pure topological" phase of the theory in which all correlation functions are determined by recursion relations. We also examine macroscopic loop amplitudes, which suggest a relation to 2D gravity coupled to dense polymers.

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2 Method

where math expression , math expression , and math expression are matrices satisfying math expression . Finally, redefining math expression we obtain (dropping title from math expression) math expression = 4(D+q) _3 A . math expression We will need the resolvent math expression R 1 (D+q) _3 . 4 . math expression In their work on the resolvents of matrix differential operators \ Gelland and Diki show that math expression satisfies

$$L(\theta) = \sum_{i=1}^n l(x_i, \theta) \tag{1}$$

Equation 1 is included to keep the sample paper-like while remaining small.

3 Conclusion

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1

PDF page-tree count; no rasterization

A one-line structural check catches what every page metric misses

Page count · higher is better

REFERENCE · complete document

Reference: 2 page(s)

p1

p2

CANDIDATE · complete document

Candidate A: 5 page(s)

p1

p2

p3

p4

p5

FIVE CONTROLLED OUTPUTS

bar = relative penalty response within this pilot

E1	Prose shift		1.000
E2	Math blur		1.000
E3	Table deletion		1.000
E4	Figure block		1.000
E5	Extra pages		0.400

It is perfect on the four page-level failures—and uniquely drops the five-page candidate to 0.400.

Sequence alignment evaluates the whole ordered document

Globally align ordered page thumbnails, permitting gaps for missing or inserted pages.

Sequence score = best ordered page match / max page count

MEASURES	Page appearance, order, insertion, and deletion
USEFUL WHEN	Multi-page document flow
BLIND SPOT	Fine semantic errors inside otherwise similar pages

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Needleman-Wunsch-style dynamic program; page SSIM matches; zero gaps

Page alignment catches extra pages without discarding quality

Page sequence · higher is better

REFERENCE · complete document

Reference: 2 page(s)

p1

p2

CANDIDATE · complete document

Candidate A: 5 page(s)

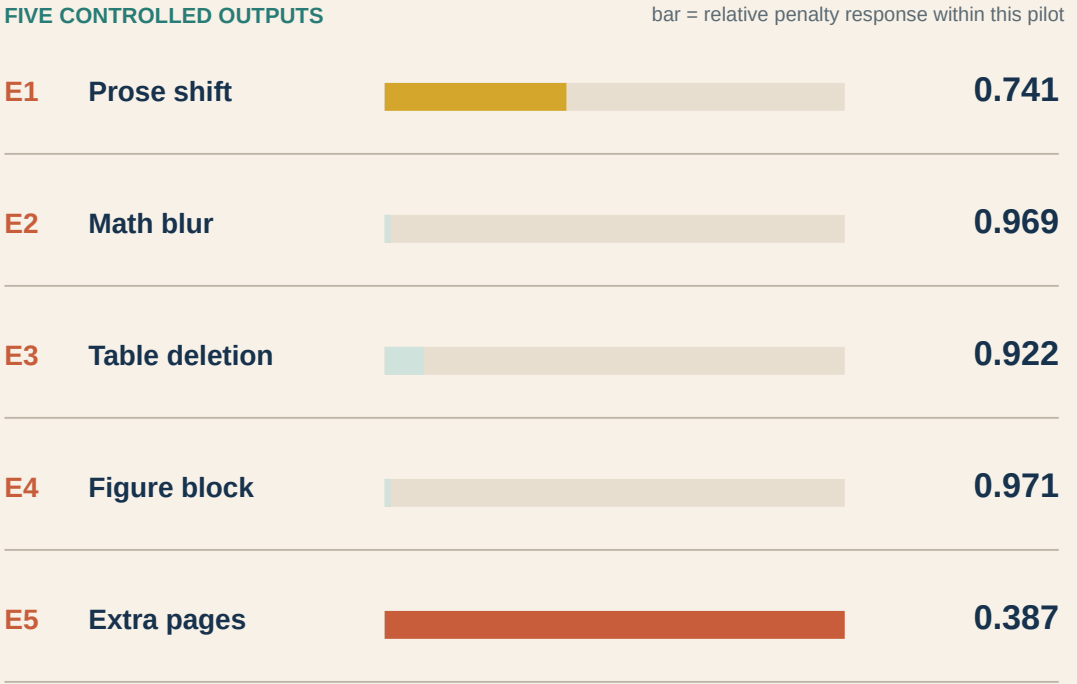
p1

p2

p3

p4

p5



Extra pages fall to 0.387 while the same target page scores nearly perfect under page-only metrics.

No single method covers all five failure modes

Cells show relative response within each method: strong, partial, or weak.

	E1 Prose shift	E2 Math blur	E3 Table deletion	E4 Figure block	E5 Extra pages
01 MAE	strong	partial	weak	weak	weak
02 PSNR	strong	strong	partial	strong	weak
03 UQI	strong	weak	weak	partial	weak
04 SSIM	strong	partial	weak	weak	weak
05 MS-SSIM	strong	weak	weak	weak	weak
06 VIF	strong	strong	weak	weak	weak
07 GMSD	strong	partial	weak	weak	weak
08 Ink F1	strong	weak	weak	weak	weak
09 Chamfer	strong	partial	weak	weak	weak
10 CER	weak	partial	strong	weak	weak
11 WER	weak	strong	strong	weak	weak
12 Token F1	weak	strong	strong	weak	weak
13 DocSim	partial	strong	weak	weak	weak
14 Word IoU	strong	strong	strong	weak	weak
15 Page count	weak	weak	weak	weak	strong
16 Page sequence	partial	weak	weak	weak	strong

The diagonal is the point: different evidence channels dominate different failures.

A defensible evaluator is a gated five-axis scorecard

CONTENT	CER · WER · token F1	Missing text or wrong symbols
LAYOUT	Word IoU · DocSim · chamfer	Where regions and lines moved
TYPOGRAPHY	VIF · GMSD · MS-SSIM	Stroke and rendering degradation
OBJECTS	Ink overlap + region metrics	Figures, tables, formulas
PAGINATION	Page count · sequence alignment	Missing, inserted, reordered pages

Critical content, object, or pagination failures gate the result. No weighted percentage until human calibration.

Human calibration should decide thresholds—not intuition

01

Expand controlled failures

Include reflow, wrong digits, font substitution, table structure, crop, reorder, and missing pages.

02

Collect simple pairwise judgments

Ask only which PDF is more faithful; keep metric values and perturbation labels hidden.

03

Fit per-axis mappings

Measure pairwise accuracy, monotonicity, uncertainty, and critical-error recall on development data.

04

Freeze, then test held-out

Do not use prompt-development samples for final benchmark claims.

Success means the score orders obvious degradations correctly, catches critical failures, and can explain every penalty.

Choose the failure model before choosing the metric.

Pixel metrics, OCR metrics, layout matching, and document structure are complementary evidence—not interchangeable percentages.

16 methods · 5 controlled examples · 1 clear conclusion